

ABHIMITHRA PEDDI

AI / Machine Learning Engineer — Generative AI, LLMs & Agentic Systems

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PROFESSIONAL SUMMARY

AI/ML Engineer with 7+ years of total professional experience, including 5+ years of relevant IT experience in machine learning, deep learning, and Generative AI. The other 2+ years were in client-facing recruitment for US and Canadian clients. My career has moved from data science and forecasting models to owning production GenAI systems end to end. Currently builds agentic AI workflows in which LLM agents call tools through function calling to automate website QA at scale. Also built a RAG document intelligence system (LangChain, Pinecone, hybrid vector + BM25 retrieval) that cut manual document search time by 80%+. Owns both the models and the infrastructure behind them, including:

- SageMaker retraining pipelines
- AWS Lambda, EC2, S3, and RDS
- IAM and encryption
- Flask REST APIs
- Run logging that makes every automated execution traceable

TECHNICAL SKILLS

- **Generative AI & LLMs:** LLM Application Development, RAG, LangChain, OpenAI API, Embeddings, Chunking Strategies, Hybrid Search (Vector + BM25), Prompt Engineering, Structured Output Parsing, Token Management, Dynamic Prompt Truncation, Response Caching
- **Agentic AI:** AI Agents, Agentic Workflows, Tool/Function Calling, Multi-Step Task Orchestration, Human-in-the-Loop Review, Rule-Based Fallback Logic, Automated Validation Pipelines
- **ML & Deep Learning:** PyTorch, TensorFlow, Keras, Scikit-learn, NLP, CNNs, RNNs, Computer Vision (OpenCV, MediaPipe), Time-Series Forecasting, Predictive Modelling, Feature Engineering, Hyperparameter Tuning
- **Cloud, MLOps & Deployment (AWS):** SageMaker Pipelines, Model Retraining & Drift Handling, EC2, Lambda, S3, RDS, IAM, Encryption & Access Control, Lex, Rekognition, Polly, Docker, CI/CD
- **APIs & Backend:** Python (Advanced), REST API Design, Flask, Django, Streamlit, React.js
- **Data & Databases:** SQL, MySQL, MongoDB, Pinecone (Vector DB), Pandas, NumPy
- **Dev Tools:** Git, GitHub Copilot, ChatGPT/Claude, Jupyter, VS Code, PyCharm

PROFESSIONAL EXPERIENCE

Machine Learning Engineer | vSplash Techlabs Private Limited, Hyderabad

Sep 2023 – Sep 2026

- Architected and built an agentic AI platform that replaced page-by-page manual website QA across client sites. LLM agents run multi-step validation workflows, calling SEO, broken-link, content-quality, and accessibility tools through function calling. A deterministic rule-based fallback takes over when model confidence is low.
- Designed a meta-model validation layer that rolls per-page scores into a site-level quality index and routes flagged issues to human reviewers, substantially reducing back-and-forth in QA cycles.
- Built a RAG document intelligence chatbot over enterprise PDFs using LangChain, Pinecone, and hybrid vector + BM25 retrieval, cutting manual document search time by 80%+.
- Applied NLP and LLM classification to content-processing pipelines, improving processing efficiency by ~25% and reducing manual intervention.
- Designed prompt templates with strict output parsing so the LLM reliably returns correctly structured SEO metadata and content schemas, keeping hallucinated output out of downstream pipelines.
- Reduced LLM latency and per-request API cost for bulk content generation through token management, dynamic prompt truncation, and response caching.
- Built and deployed Flask REST APIs for secure model serving and real-time automation across multiple client websites.
- Owned AWS deployment and operations on EC2, Lambda, S3, and RDS. Ran SageMaker pipelines that retrain models as input data drifts, integrated Lex, Rekognition, and Polly, and configured IAM roles and encryption for security and compliance.
- Logged every QA result and automated run to MongoDB, giving full traceability for client audits and faster debugging of model regressions.

Data Scientist | Young Minds Technology Solutions Pvt. Ltd., Tirupati

Sep 2021 – Aug 2023

- Built predictive and deep learning forecasting models end to end, from raw data through production deployment.
- Built Flask and Django applications that let business teams use model outputs directly, removing analyst bottlenecks.
- Extended into computer vision, using OpenCV to extract structured data from images in production.

- Maintained and optimised the SQL databases feeding Python pipelines, materially improving the reliability of downstream analytics.

Data Science Intern | Innomatics Research Labs, Hyderabad

Jan 2021 – Apr 2021

- Ran EDA and feature engineering on large financial datasets to prepare them for production-grade modelling.
- Built loan-decision models and refined them iteratively until precision reached a client-presentable standard.

HR Associate | Tenpath Solutions, Hyderabad

Aug 2018 – Nov 2020

- Managed end-to-end recruitment for US and Canadian clients across technical and non-technical roles.
- Owned US client accounts, gathering hiring requirements and coordinating between hiring managers and candidates to close positions on time. Built stakeholder-management skills that carried into scoping ML projects.

KEY PROJECTS

- **Agentic QA Automation Platform** — *Python, Flask, LangChain, OpenAI, AWS (EC2, Lambda, S3, RDS, SageMaker), MongoDB*
Production agentic AI platform that automates website QA at scale. LLM agents orchestrate multi-step validation workflows through function calling. A meta-model aggregates per-page scores into a site-level quality index and routes anomalies to human reviewers, while rule-based fallback keeps results deterministic when LLM confidence is low. Flask REST APIs serve real-time results to client dashboards, and MongoDB audit logs make every run traceable.
- **SM Dashboard** — *Python, MySQL, SLA & Billing Management*
Built an SM Dashboard in Python and MySQL to monitor SLA performance across multiple SM products. It tracks order assignment, manager ownership, processing status, and SLA timelines across product teams. Managers can log delay comments and reasons and set revised SLA timelines, while an integrated finance approval workflow tracks billing requests through approval or rejection. Dashboard views for order tracking, SLA compliance, manager actions, and billing status improved operational visibility and accountability.
- **Document Intelligence Chatbot (RAG)** — *LangChain, Pinecone, Python, OpenAI Embeddings*
RAG system over enterprise PDFs that cut manual search time by 80%+. It uses Pinecone hybrid retrieval (vector + BM25), RecursiveCharacterTextSplitter chunking, and prompt engineering tuned for context handling.
- **COVID-19 Detection from Chest X-Rays** — *Python, TensorFlow, Keras (VGG16 / VGG19 / ResNet50)*
Trained and benchmarked three CNN architectures on doctor-screened X-ray data, measuring how preprocessing choices affected classification performance.
- **Sentiment Analysis on Movie Reviews** — *Python, RNN, NLP*
Built an RNN classifier that handled negation and mid-sentence context shifts noticeably better than classical ML baselines.
- **SVM Model Deployment** — *Scikit-learn, Streamlit, Heroku*
Took an SVM model from notebook to a live web application.
- **Body Posture Controller** — *OpenCV, MediaPipe, Tkinter*
Translated real-time body-landmark tracking into character movement controls.
- **CO2 Emission Forecasting** — *Python, Statsmodels, ARIMA*
Forecast monthly CO2 emissions with Seasonal ARIMA, validating stationarity and seasonal decomposition throughout.

EDUCATION

- PG Diploma in Data Science (Prodegree) — Imarticus Learning
- B.Tech, Electrical and Electronics Engineering — DMS Sri Venkateswara Hindu College of Engineering, 2018

CERTIFICATIONS & ACHIEVEMENTS

- IBM Certified Specialist — IBM SPSS Modeller Level 1
- Ranked 473rd of 10,000+ participants — Analytics Vidhya Hackathon
- Paper Presentation — Sri Vasavi College of Engineering Fest